ACI 318-19 RC-SL EXAMPLE 001

Slab Flexural Design

PROBLEM DESCRIPTION

The purpose of this example is to verify slab flexural design in ETABS and SAFE.

A one-way, simple-span slab supported by walls on two opposite edges is modeled. The slab is 6 inches thick and spans 12 feet between walls. The walls are modeled as line supports. The computational model uses a finite element mesh, automatically generated by ETABS, SAFE. The maximum element size is specified to be 36 inches. To obtain factored moments and flexural reinforcement in a design strip, one one-foot-wide strip is defined in the X-direction on the slab, as shown in Figure 1.

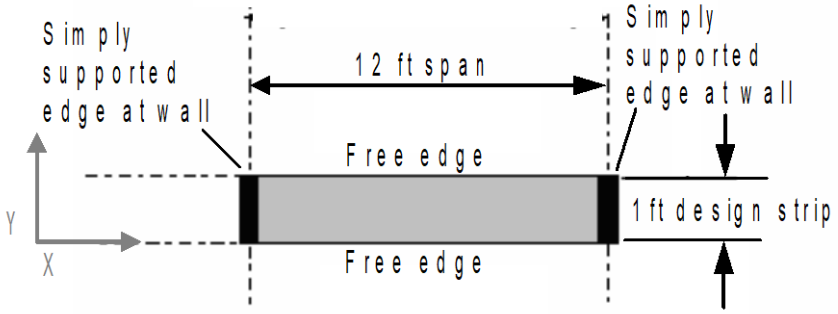


Figure 1 Plan View of One-Way Slab

One dead load case (DL80) and one live load case (LL100) with uniformly distributed surface loads of magnitudes 80 and 100 psf, respectively, are defined in the model. A load combination (COMB100) is defined using the ACI 318-19 load combination factors, 1.2 for dead loads and 1.6 for live loads. The model is analyzed for both load cases and the load combination.

The slab moment on a strip of unit width is computed analytically. The total factored strip moments are compared with the results. After completing the analysis, design is performed in accordance with ACI 318-19 and also by hand computation. Table 1 shows the comparison of the moments and design reinforcements computed using the two methods.

PROGRAM NAME: ETABS, SAFE
REVISION NO.: 1

GEOMETRY, PROPERTIES AND LOADING

Thickness	T, h	=	6	in
Depth of tensile reinf.	d_c	=	1	in
Effective depth	d	=	5	in
Clear span	l_n, l_l	=	144	in
Concrete strength	f_c	=	4,000	psi
Yield strength of steel	f_y	=	60,000	psi
Concrete unit weight	w_c	=	0	pcf
Modulus of elasticity	E_c	=	3,600	ksi
Modulus of elasticity	E_s	=	29,000	ksi
Poisson's ratio	ν	=	0	
Dead load	w_d	=	80	psf
Live load	w_l	=	100	psf

TECHNICAL FEATURES TESTED

- Calculation of flexural reinforcement
- Application of minimum flexural reinforcement

RESULTS COMPARISON

Table 1 shows the comparison of the total factored moments in the design strip with the moments obtained by the hand computation method. Table 1 also shows the comparison of the design reinforcements.

Table 1 Comparison of Design Moments and Reinforcements

Load Level	Method	Strip Moment (k-in)	Reinforcement Area (sq-in)
			A_s^+
Medium	ETABS, SAFE	55.22	0.213
	Calculated	55.22	0.213

$A^+_{s,\min}$ = 0.1296 sq-in

COMPUTER FILE: ACI 318-19 RC-SL Ex001

CONCLUSION

The results show an acceptable comparison with the independent results.

HAND CALCULATION

The following quantities are computed for the load combination:

$\varphi = 0.9$

$b = 12$ in

$A_{s,min} = 0.0018bh = 0.1296$ sq-in

$$\beta_1 = 0.85 - 0.05\left(\frac{f_c' - 4000}{1000}\right) = 0.85$$

$$c_{max} = \frac{0.003}{0.003 + \frac{f_y}{E}} d = 2.95 \text{ in}$$

$a_{max} = \beta_1 c_{max} = 2.5$ in

For the load combination, w and M_u are calculated as follows:

$w = (1.2w_d + 1.6w_t)\, b / 144$

$$M_u = \frac{wl_1^2}{8}$$

COMB100

$w_d = 80$ psf

$w_t = 100$ psf

$w = 21.33$ lb/in

$M_{u\text{-}strip} = 55.22$ k-in

$M_{u\text{-}design} = 55.629$ k-in

The depth of the compression block is given by:

$$a = d - \sqrt{d^2 - \frac{2|M_u|}{0.85 f_c' \varphi\, b}} = 0.3128 \text{ in} < a_{max}$$

The area of tensile steel reinforcement is then given by:

$$A_s = \frac{M_u}{\varphi f_y\left(d - \frac{a}{2}\right)} = 0.213 \text{ sq-in} > A_{s,min}$$

$A_s = 0.213$ sq-in